# PROJECT AND TEAM INFORMATION

## Project Title

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| *Cyber Crime Case Management System* |

## Student / Team Information

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| *Team Name:*<br>*Team #* | *The Solve Squad* |
| **Team member 1 (Team Lead)**<br>*(Sapra, Kusum: 2510011573: 2510011573@geu.ac.in):* | *Sapra, Kusum– 2510011573 2510011573@geu.ac.in*<br>  |
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# PROPOSAL DESCRIPTION (10 pts)

## Motivation (1 pt)

*Everything nowdays is online banking,shopping,communication and other day to day activities.dependency of doing things online has increased due to which cyber crime also increased.fraud,phishing,online harassment has increased and it needs to be properly organised. The motivation for this project is to organise cyber crime in one place and proper order.this system will make the work easier to search cases,track updates and follow up ongoing cases.it is an application of data structures and object oriented programming. Different data structures will be used.Linked list will be used for managing data,queues will be used for handling case and priorities,stack for maintaining investigation history and Hash table for searching.c++ object oriented programming will be used to organise the code making the system easier to work.*

## State of the Art / Current solution (1 pt)

*Nowadays, cybercrime cases can involve a lot of information, such as case details, evidence, investigation updates, reports, and activity records. Keeping all the information can be difficult when there are many cases at the same time. That is why different organizations use computer-based systems to keep records tracked.Professional systems used by police departments and government organizations can store case information, manage evidence, track investigation progress, control access, and prepare reports. Some advanced systems may also use technologies such as blockchain to make digital evidence more secure and easier to track. These systems are designed for large organizations and contain many advance features. Some advanced systems may also use technologies such as blockchain to make digital evidence more secure and easier to track.*
*Our idea is to create a basic cybercrime case management system where all the important case information can be stored in one place. We will use different data structures for different tasks. For example, linked lists can help store cases, queues can manage cases in order, priority queues can handle urgent cases, stacks can keep track of recent activities, and hash tables can make searching for cases faster.Through this project, we want to understand how the concepts we study in class can actually be used to solve a real-world type of problem. It will also help us understand why choosing the right data structure is important when developing a software system.*

## Project Goals and Milestones (2 pts)

*The main goal of the project is to develop a simple and practical system for managing cybercrime cases. The system will use different data structures to store, search, organize, and prioritize case information efficiently. C++ Object-Oriented Programming will be used to keep the software structured, maintainable, and easier to expand.*
*The system will focus on tracking important case details, investigation updates, evidence information, case status, and investigation history. Another important goal is to gain practical understanding of how different data structures perform when used for different tasks rather than only studying their theoretical concepts.*
*The major milestones of the project include:*

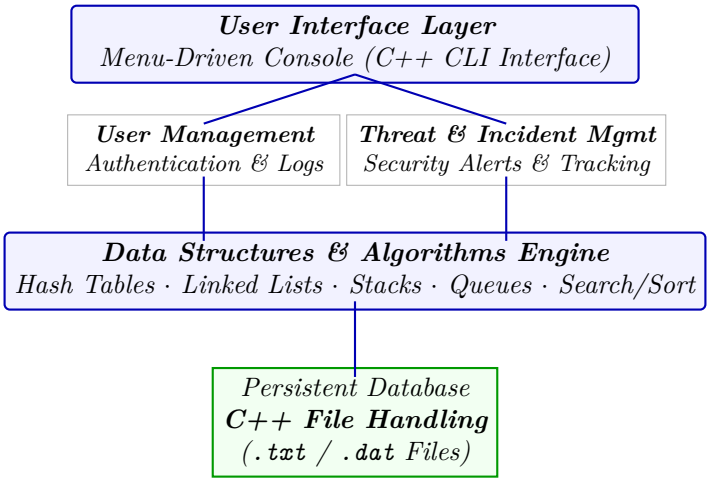
- *Requirement Analysis*
- *Data Structure Implementation*
- *OOP Development*
- *Integration of all modules*
- *Testing using sample cybercrime cases*
- *Evaluation and comparison of data structures*
- *Documentation and final project report*

*By completing these milestones, the project will demonstrate how data structures and OOP can be combined to create a functional case management system.*

## Project Approach (3 pts)

*Instead of building the entire system , this project follows an step-by-step development strategy. The initial phase focuses entirely on analyzing system needs and establishing foundational data structures to store,and manage cybercrime incident records. we will implement various data structures, including linked lists, queues, priority queues, stacks, and hash tables. Once these data structures are fully functional, we will work with C++ Object-Oriented Programming (OOP). This involves designing dedicated classes to model critical entities like incidents, officers, and evidence. We will actively apply core OOPs principles—namely encapsulation, inheritance, and polymorphism—to ensure our codebase remains clean, modular, and easy to maintain.We will also integrate basic file I/O operations, ensuring that case data remains locally on disk even after shutting down the program.Finally, we will validate the software using realistic cybercrime scenarios. This testing phase will help us hunt down bugs, confirm that different modules interface works, and measure how efficiently our chosen data structures perform under load.*

## System Architecture (High Level Diagram)(2 pts)




## Project Outcome / Deliverables (1 pts)

*The main outcome of the project will be a working system for organizing and managing cybercrime cases. The system will provide structured storage for case details, updates, evidence, investigation history, and case status.*

*The use of appropriate data structures will allow faster searching and prioritization of cases. Linked lists, queues, priority queues, stacks, and hash tables will be used according to the requirements of different operations. C++ Object-Oriented Programming will help keep the implementation organized and maintainable.*

*The system will also provide reliable tracking of case status, evidence, and investigation history. Basic local file handling will allow information to be saved and loaded when required.*

*The final deliverables will include the working C++ application, implemented data structures, OOP-based classes, file-handling functionality, sample cybercrime cases, test cases and outputs, system architecture, evaluation of data structures, and project documentation.*

*Overall, the project will provide practical hands-on experience with both C data structures and C++ programming while creating a simple system that can be extended with additional features in the future.*

## Assumptions

*The project will use sample or simulated cybercrime data rather than real criminal cases or sensitive information.We're using fake data: We are only testing this with simulated cybercrime data, not real police files or sensitive cases. Anyone using the app needs to know it's just for practice and shouldn't type in real private info.Users know the basics: We assume whoever opens the program knows how to use a computer and understands the information they are typing. Garbage in, garbage out: The program assumes whatever data the user types in is correct. It won't automatically fix formatting or catch typos for you.Everything saves locally: To keep things simple for this version, we are saving everything straight to local text files instead of setting up a cloud server or external database. Since it's a basic prototype, we plan to add actual cloud support and better features later on if we expand the project.*

## References

*M. T. Goodrich, R. Tamassia, and M. H. Goldwasser, Data Structures and Algorithms in C++, Wiley.*
*Robert Lafore, Data Structures and Algorithms in Java, Sams Publishing. (Useful for understanding data-structure concepts, even though the implementation is in C++.)*
*Bjarne Stroustrup, Programming: Principles and Practice Using C++, Addison-Wesley.*
*GeeksforGeeks – Data Structures and Algorithms*
*cppreference – C++ Language and Standard Library Reference*
*W3Schools – C++ Tutorial*
*GitHub – Code Repository and Version Control Platform*
*Microsoft Visual Studio Code – Development Environment*
*OWASP – Cybersecurity and Application Security Resources*
*NIST Cybersecurity Framework*